UC SAN DIEGO / ECE 251C

9

Designing two-channel perfect-reconstruction filter banks

Executive summary

We use the perfect-reconstruction conditions to design a two-channel filter bank. A suitable choice of determinant reduces the four-filter problem to the design and factorization of one product filter. Its polyphase representation shows that one set of alternating taps is fixed, while the remaining taps are free. This leads to half-band filters and their interpolation property. For a symmetric product, the same condition becomes a complementarity identity for the signed amplitude. We use this identity to relate the passband, stopband, and their ripples, and then examine the ideal low-pass limit.

9.1 From verification to design

In Lecture 8, we derived the input–output relation for a two-channel bank and identified the conditions for perfect reconstruction. We now use those identities as design equations. The analysis and synthesis filters cannot be chosen independently: together they must cancel aliasing and reproduce the input up to a fixed delay. Our first step is to fix the analysis filters and solve for the synthesis filters. The form of that solution will suggest how to keep all four filters finite.

We keep the same two-channel system. The analysis filters are h_0, h_1, and the synthesis filters are g_0, g_1. No coefficient processing is inserted between the banks.

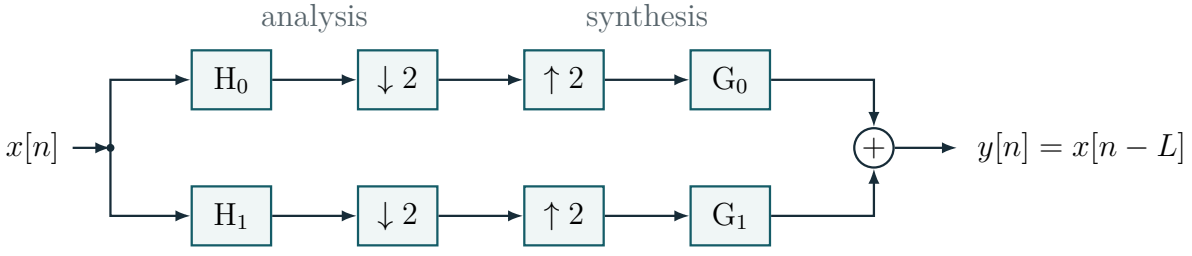


Figure 9.1: The same architecture as Lecture 8. The synthesis stage upsamples before filtering. We now design the four filters rather than verify a supplied choice.

Recall the input–output relation. With $X(z) = \sum_n x[n]z^{-n}$,

$$\begin{aligned} Y(z) = \frac{1}{2}X(z)\underbrace{\big[G_0(z)H_0(z) + G_1(z)H_1(z)\big]}_{\text{desired-signal term}} \\ + \frac{1}{2}X(-z)\underbrace{\big[G_0(z)H_0(-z) + G_1(z)H_1(-z)\big]}_{\text{alias term}}. \end{aligned} \tag{9.1}$$

Perfect reconstruction means $y[n] = x[n-L]$, or $Y(z) = z^{-L}X(z)$. The integer L is the overall delay; in a causal implementation we take $L \geq 0$.

9.2 Perfect reconstruction as a linear system

The two identities from Lecture 8 are

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2z^{-L}, \tag{9.2}$$

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0. \tag{9.3}$$

The first preserves the signal up to a delay. The second removes the shifted spectrum $X(-z)$. These are the perfect-reconstruction conditions (Vetterli, 1986).

The two identities constrain four filter responses. This leaves design freedom, but also makes it important to distinguish the choices from their consequences. One approach is to prescribe the analysis filters and regard the identities as a linear system for the two synthesis responses:

$$\begin{bmatrix} H_0(z) & H_1(z) \\ H_0(-z) & H_1(-z) \end{bmatrix} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} = \begin{bmatrix} 2z^{-L} \\ 0 \end{bmatrix}. \tag{9.4}$$

This is a 2×2 linear system whose entries are functions of z. Its determinant is

$$\Delta(z) \coloneqq H_0(z)H_1(-z) - H_0(-z)H_1(z). \tag{9.5}$$

When Δ is not identically zero, the inverse-matrix formula gives a rational solution. Wherever the denominator is nonzero,

$$\begin{aligned} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} &= \frac{1}{\Delta(z)} \begin{bmatrix} H_1(-z) & -H_1(z) \\ -H_0(-z) & H_0(z) \end{bmatrix} \begin{bmatrix} 2z^{-L} \\ 0 \end{bmatrix} \\ &= \frac{2z^{-L}}{\Delta(z)} \begin{bmatrix} H_1(-z) \\ -H_0(-z) \end{bmatrix}. \end{aligned}$$

In particular,

$$\boxed{G_0(z) = \frac{2z^{-L}H_1(-z)}{\Delta(z)}, \qquad G_1(z) = -\frac{2z^{-L}H_0(-z)}{\Delta(z)}.} \tag{9.6}$$

The entries of the inverse are obtained by interchanging the diagonal entries and negating the off-diagonal entries. Multiplication by the right-hand side then leaves the first column, scaled by $2z^{-L}$.

The inverse formula solves the algebraic reconstruction problem. To use it for filter design, we must also examine the denominator: a rational expression need not describe a finite-length filter, and inversion by itself does not guarantee a stable causal implementation.

9.3 Keeping the synthesis filters finite

An FIR filter has finitely many nonzero coefficients, so its z-transform is a finite sum of integer powers of z. Multiplication by a delay z^{-r} only shifts those coefficients. Division by a nontrivial polynomial, however, can create an infinite impulse response.

When H_0 and H_1 are FIR, the numerators in equation (9.6) are finite sums, but the determinant in the denominator need not cancel. The following example shows how an infinite synthesis filter can arise from finite analysis filters.

A short example of the denominator problem. Take $H_0(z) = 1 - \frac{1}{2}z^{-2}$, $H_1(z) = z^{-1}$, and $L = 1$. Since $H_0(-z) = H_0(z)$ and $H_1(-z) = -z^{-1}$,

$$\Delta(z) = -2z^{-1}\big(1 - \tfrac{1}{2}z^{-2}\big),$$
$$G_0(z) = \frac{z^{-1}}{1 - \frac{1}{2}z^{-2}} = \sum_{m=0}^{\infty} 2^{-m} z^{-(2m+1)}, \qquad G_1(z) = 1.$$

The geometric expansion converges for $|z| > 1/\sqrt{2}$, which includes the unit circle. Thus $g_0[2m+1] = 2^{-m}$ for $m \geq 0$, with all even taps zero. This is a stable causal synthesis filter, but it is not FIR. The example illustrates the denominator issue; it is not intended as a frequency-selective low-pass/high-pass design.

A determinant that cannot create a tail. Choose

$$\boxed{\Delta(z) = 2z^{-L}.} \tag{9.7}$$

The common factor in equation (9.6) then cancels, leaving

$$\boxed{G_0(z) = H_1(-z), \qquad G_1(z) = -H_0(-z).} \tag{9.8}$$

Replacing z by $-z$ alternates the coefficient signs:

$$H(-z) = \sum_n (-1)^n h[n] z^{-n}.$$

It does not change the support. Consequently the filters in equation (9.8) are FIR whenever H_0, H_1 are FIR.

More generally, choosing the determinant to be a nonzero constant times a delay avoids division by a nontrivial polynomial. The normalization in equation (9.7) is useful because it gives the simple completion rule above. Other placements of delays and gains lead to other, equally valid, perfect-reconstruction conventions.

9.4 Replace four filters by one product

Under equation (9.8), define the *product filter*

$$P_0(z) := H_0(z)G_0(z), \qquad p_0 = h_0 * g_0. \tag{9.9}$$

Since $H_1(z) = G_0(-z)$, the determinant becomes

$$\begin{aligned}\Delta(z) &= H_0(z)G_0(z) - H_0(-z)G_0(-z)\\ &= P_0(z) - P_0(-z).\end{aligned}$$

The four-filter design problem has therefore reduced to a single identity:

$$\boxed{P_0(z) - P_0(-z) = 2z^{-L}.} \tag{9.10}$$

The design procedure. First choose an FIR P_0 satisfying equation (9.10). Next factor it as $P_0 = H_0G_0$, with FIR factors. Finally set

$$H_1(z) = G_0(-z), \qquad G_1(z) = -H_0(-z). \tag{9.11}$$

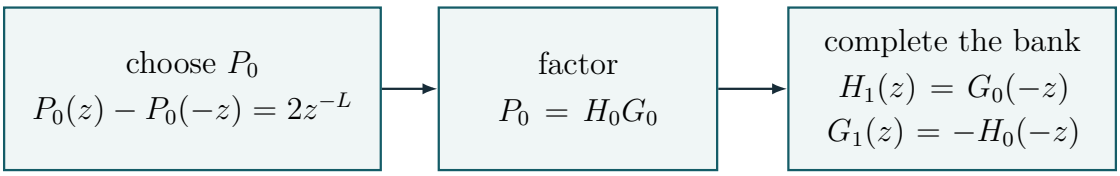


Figure 9.2: The product identity handles reconstruction. The factorization determines how that product is divided between analysis and synthesis.

Proposition 9.1. The preceding construction gives perfect reconstruction with delay L.

Proof. The alias expression cancels identically:

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = G_0(z)H_0(-z) - H_0(-z)G_0(z) = 0.$$

The desired expression is

$$G_0(z)H_0(z) + G_1(z)H_1(z) = P_0(z) - P_0(-z) = 2z^{-L}.$$

Apply equation (9.1) to obtain $Y(z) = z^{-L}X(z)$. □

The product fixes the reconstruction identity, but it does not uniquely determine its factors. Different factorizations distribute support, symmetry, gain, and frequency selectivity differently between analysis and synthesis. This freedom will be central when we design wavelet filters. Perfect reconstruction alone does not yet guarantee that those filters generate an orthonormal wavelet basis.

9.5 Polyphase separates even and odd taps

The factorization step can wait until we have a suitable product. To design P_0, we first need to understand what changes when z is replaced by $-z$. The answer is simplest when we separate the even-indexed and odd-indexed coefficients. This is the *polyphase representation*.

For a filter h, define two sequences on the coarser index grid:

$$h_{\text{even}}[m] := h[2m], \qquad h_{\text{odd}}[m] := h[2m+1].$$

This construction separates the coefficients according to the parity of their indices. The terms "even" and "odd" do not refer to symmetry of the sequence as a function. No coefficient has been altered or discarded; the same taps have simply been placed on two coarser grids.

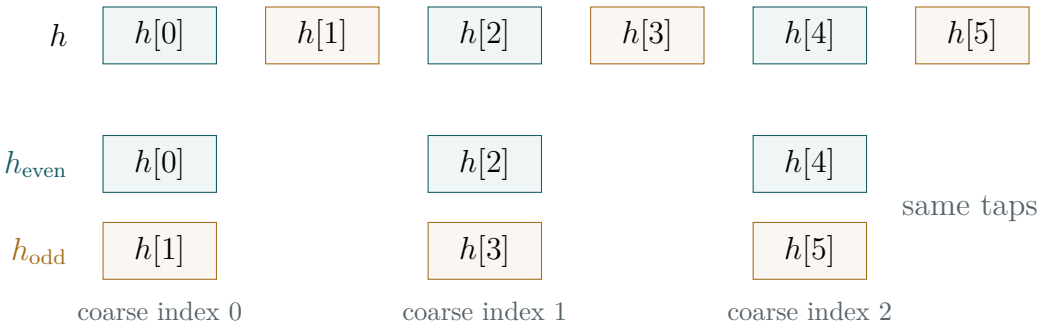


Figure 9.3: Even and odd locations become two sequences, each indexed by m. Interleaving them recovers the original filter exactly.

Separate the defining sum for $H(z)$ into its two index classes:

$$\begin{aligned} H(z) &= \sum_m h[2m] z^{-2m} + \sum_m h[2m+1] z^{-(2m+1)} \\ &= \sum_m h_{\text{even}}[m] (z^2)^{-m} + z^{-1} \sum_m h_{\text{odd}}[m] (z^2)^{-m} \\ &= H_{\text{even}}(z^2) + z^{-1} H_{\text{odd}}(z^2). \end{aligned} \tag{9.12}$$

The substitution z^2 inserts a zero between successive coefficients. The additional z^{-1} moves the odd branch onto the odd locations. In sequence notation,

$$h[n] = (\uparrow 2\, h_{\text{even}})[n] + (\uparrow 2\, h_{\text{odd}})[n-1].$$

For $n = 2m$, only the first term survives; for $n = 2m+1$, only the second survives. This checks every index in the reconstruction.

Polyphase supports efficient interpolation and antialias filtering followed by downsampling, as well as filter-bank analysis. Here its immediate purpose is algebraic: it identifies which product-filter coefficients are fixed by perfect reconstruction.

9.6 The odd component must be a delay

Apply equation (9.12) to P_0:

$$\begin{aligned} P_0(z) &= P_{0,\text{even}}(z^2) + z^{-1} P_{0,\text{odd}}(z^2), \\ P_0(-z) &= P_{0,\text{even}}(z^2) - z^{-1} P_{0,\text{odd}}(z^2). \end{aligned}$$

Subtracting removes all even-indexed coefficients. Thus equation (9.10) is equivalent to

$$2z^{-1} P_{0,\text{odd}}(z^2) = 2z^{-L}. \tag{9.13}$$

Every power on the left has an odd exponent. Therefore L must be odd. Write

$$L = 2k+1, \qquad k \in \mathbb{Z}.$$

After canceling $2z^{-1}$ in equation (9.13),

$$P_{0,\text{odd}}(z^2) = z^{-2k} = (z^2)^{-k}.$$

Equating coefficients, or renaming the transform variable, gives

$$\boxed{P_{0,\text{odd}}(z) = z^{-k}, \qquad p_0[2m+1] = \delta[m-k].} \tag{9.14}$$

There is exactly one nonzero odd-indexed coefficient: $p_0[L] = 1$. The even-indexed coefficients are the design variables.

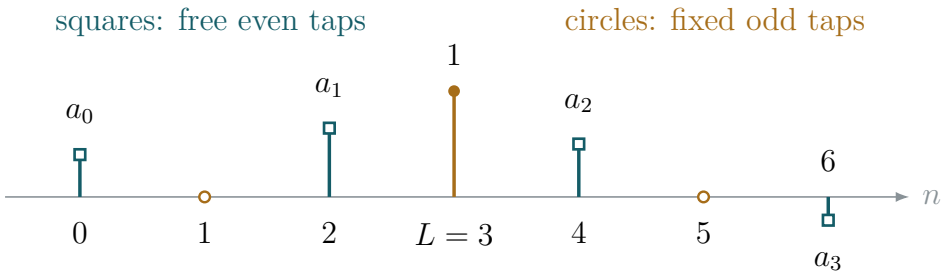


Figure 9.4: The coefficient pattern for $L = 3$. The heights of the square-marked taps are schematic design choices; the odd grid has one unit tap and zeros elsewhere.

Equivalently, we may choose any FIR even component $E(z)$ and form

$$P_0(z) = E(z^2) + z^{-L}, \qquad L \text{ odd}. \tag{9.15}$$

This coefficient pattern guarantees the product identity for every $z \neq 0$. It does not yet impose a useful frequency response. Low-pass behavior, symmetry, and a convenient factorization must be imposed through the remaining even-indexed coefficients.

Why this does not contradict the previous lecture. The odd-delay conclusion belongs to the particular normalization equations (9.7) and (9.11). It is not a restriction on every PR bank. The zero-delay Haar bank from Lecture 8 used a different placement of shifts and signs.

9.7 A three-tap product and its factorization

The first example has $(p_0[0], p_0[1], p_0[2]) = (\frac{1}{2}, 1, \frac{1}{2})$, with all other taps zero. The product and its polyphase components are

$$P_0(z) = \tfrac{1}{2} + z^{-1} + \tfrac{1}{2}z^{-2},$$
$$P_{0,\text{even}}(z) = \tfrac{1}{2} + \tfrac{1}{2}z^{-1}, \qquad P_{0,\text{odd}}(z) = 1.$$

Thus $k = 0$ and $L = 1$. Direct subtraction also gives $P_0(z) - P_0(-z) = 2z^{-1}$.

Complete the design. The product condition is only the first step. To obtain the four filters, factor the product as

$$P_0(z) = \tfrac{1}{2}(1 + z^{-1})^2.$$

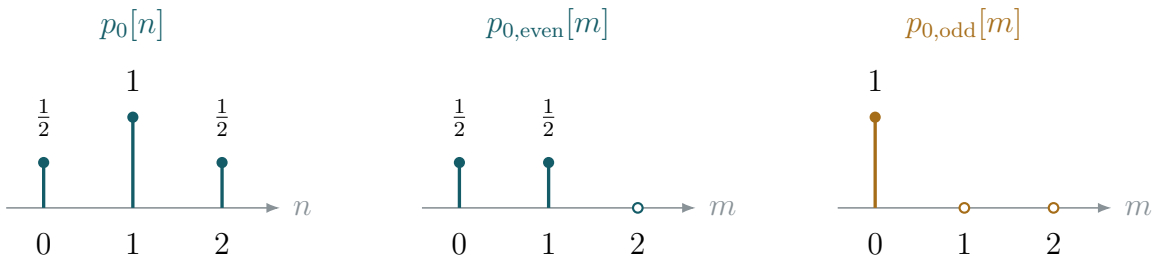


Figure 9.5: The three-tap product splits into a two-tap even component and a unit impulse on the odd grid. The latter gives $L = 2(0) + 1 = 1$.

Split the constant equally between the factors:

$$\begin{aligned} H_0(z) &= \frac{1 + z^{-1}}{\sqrt{2}}, \quad G_0(z) = \frac{1 + z^{-1}}{\sqrt{2}}, \\ H_1(z) &= \frac{1 - z^{-1}}{\sqrt{2}}, \quad G_1(z) = \frac{-1 + z^{-1}}{\sqrt{2}}. \end{aligned} \tag{9.16}$$

The second row follows from equation (9.11). These are Haar sum-and-difference filters with a causal, one-sample-delay convention. In particular, the synthesis high-pass sign is part of the construction, not a typographical choice.

Check the reconstruction on samples. If the two downsampled outputs are $c_0[m]$ and $c_1[m]$, then

$$c_0[m] = \frac{x[2m] + x[2m-1]}{\sqrt{2}}, \qquad c_1[m] = \frac{x[2m] - x[2m-1]}{\sqrt{2}}.$$

Synthesis gives

$$y[2m] = \frac{c_0[m] - c_1[m]}{\sqrt{2}} = x[2m-1], \qquad y[2m+1] = \frac{c_0[m] + c_1[m]}{\sqrt{2}} = x[2m].$$

Both index classes satisfy $y[n] = x[n-1]$.

9.8 A seven-tap product

The second example has seven taps:

$$(p_0[0], \dots, p_0[6]) = \left(-\tfrac{1}{4}, 0, \tfrac{1}{2}, 1, \tfrac{1}{2}, 0, -\tfrac{1}{4}\right).$$

Its symmetry center is $L = 3$, and its polyphase components are

$$\begin{aligned} P_{0,\text{even}}(z) &= -\tfrac{1}{4} + \tfrac{1}{2}z^{-1} + \tfrac{1}{2}z^{-2} - \tfrac{1}{4}z^{-3}, \\ P_{0,\text{odd}}(z) &= z^{-1}. \end{aligned}$$

The odd branch is again a pure delay, now with $k = 1$. Therefore

$$P_0(z) - P_0(-z) = 2z^{-3}.$$

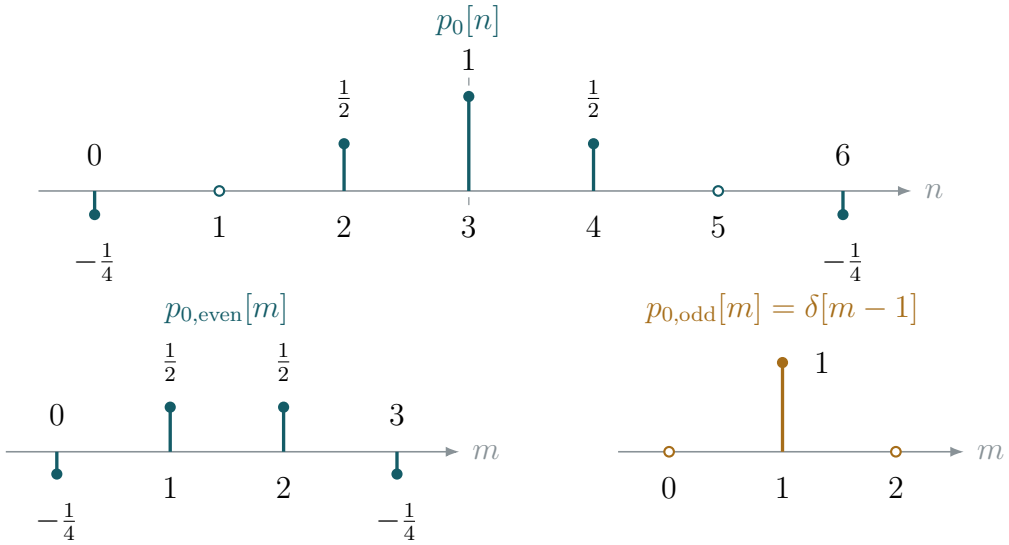


Figure 9.6: The seven-tap product and its polyphase components. The odd component has one delay, giving $L = 3$.

Again the odd grid contains one unit tap and zeros elsewhere. The even coefficients may be positive or negative. Symmetry makes this seven-tap product Type I, as in Lecture 1.

Reconstruction and frequency selectivity. The preceding calculation verifies the product identity. It does not verify a low-pass specification. Indeed, the endpoint responses are

$$P_0(1) = \sum_n p_0[n] = \frac{3}{2}, \qquad P_0(-1) = \sum_n (-1)^n p_0[n] = -\frac{1}{2}.$$

Their difference is two, as required. But the Nyquist response is not zero, and the DC response is not two. This is a valid *algebraic* half-band product, not an ideal low-pass response: reconstruction and frequency selectivity are distinct requirements.

9.9 Half-band filters and interpolation

The two examples have different lengths and different even-indexed coefficients, but they satisfy the same condition at every other sample. The rule in equation (9.14) can be written without explicitly mentioning the odd grid. Since $L = 2k + 1$,

$$p_0[2n + L] = p_0[2(n + k) + 1] = \delta[n].$$

Definition. In our convention, a filter is *half-band centered at L* when

$$\boxed{p_0[2n+L] = \delta[n], \qquad n \in \mathbb{Z}.} \tag{9.17}$$

For our design, L is odd. The definition fixes one parity of the taps. The term "half-band" is often also used with a low-pass frequency specification; that extra goal will be imposed separately below.

Why this is an interpolation condition. Upsample a sequence x by two and filter it with p_0:

$$u[n] = \begin{cases} x[n/2], & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \qquad r[n] = (p_0 * u)[n] = \sum_m x[m] p_0[n-2m].$$

At the locations $n = 2\ell + L$, the output is

$$\begin{aligned} r[2\ell+L] &= \sum_m x[m] p_0[2(\ell-m)+L] \\ &= \sum_m x[m]\delta[\ell-m] = x[\ell]. \end{aligned}$$

The original samples are therefore reproduced exactly at one parity of the output locations, with the prescribed delay. The remaining taps determine the values between those locations. This is why the same coefficient condition that appears in filter-bank reconstruction also appears in interpolation.

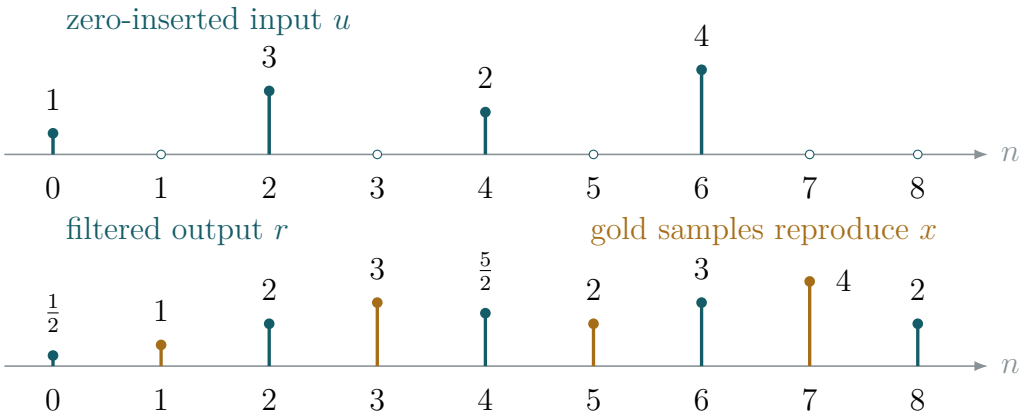


Figure 9.7: For the three-tap product, the input $(1, 3, 2, 4)$, zero elsewhere, is preserved at output indices $1, 3, 5, 7$. The intervening values are neighboring-sample averages.

Shifting the center to zero, $h[n] = p_0[n+L]$ gives the usual interpolation condition $h[2n] = \delta[n]$. Only the location of the preserved sample phase changes.

9.10 Type-I symmetry separates delay from amplitude

The product identity does not require symmetry. To obtain a linear-phase product, we now *add* the real-valued symmetry condition

$$p_0[L+r] = p_0[L-r].$$

Take a causal product supported in $\{0, \dots, 2L\}$. It has odd length and integer center L, so it is Type I. A Type-II symmetric filter has a half-integer center and does not have this centering. Here the center is the same odd integer L that appears in the product identity.

Pair equal coefficients on opposite sides of the center. For $1 \le r \le L$,

$$\begin{aligned} &p_0[L-r]\mathrm{e}^{-\mathrm{j}(L-r)\omega} + p_0[L+r]\mathrm{e}^{-\mathrm{j}(L+r)\omega} \\ &\quad = \mathrm{e}^{-\mathrm{j}L\omega} p_0[L-r]\big(\mathrm{e}^{\mathrm{j}r\omega} + \mathrm{e}^{-\mathrm{j}r\omega}\big) \\ &\quad = \mathrm{e}^{-\mathrm{j}L\omega}\, 2p_0[L-r]\cos(r\omega). \end{aligned}$$

Adding the center tap gives

$$\boxed{P_0(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}L\omega} P_{\mathrm{ampl}}(\omega),} \tag{9.18}$$

where

$$P_{\mathrm{ampl}}(\omega) = p_0[L] + 2\sum_{r=1}^{L} p_0[L-r]\cos(r\omega). \tag{9.19}$$

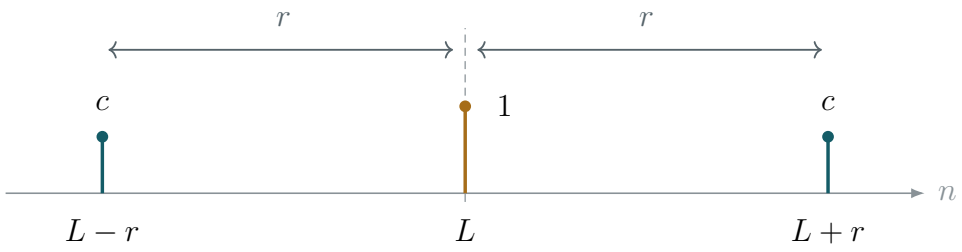


Figure 9.8: One symmetric tap pair contributes $2c\cos(r\omega)$ after the common delay $\mathrm{e}^{-\mathrm{j}L\omega}$ is removed. The center tap contributes one.

The function P_{ampl} is real, even, and 2π-periodic. It is a *signed amplitude*, not necessarily a magnitude:

$$|P_0(\mathrm{e}^{\mathrm{j}\omega})| = |P_{\mathrm{ampl}}(\omega)|.$$

A negative value of the signed amplitude corresponds to an additional phase change of π, not a negative magnitude. We must retain this sign when applying the reconstruction identity: taking absolute values would remove the cancellation between paired passband and stopband errors.

For the three-tap example, $P_{\mathrm{ampl}}(\omega) = 1 + \cos\omega$. For the seven-tap example, it is $1 + \cos\omega - \frac{1}{2}\cos(3\omega)$. These follow directly by pairing the displayed taps.

9.11 The two amplitudes must add to two

On the unit circle, replacing z by $-z$ shifts frequency by π:

$$-\mathrm{e}^{\mathrm{j}\omega} = \mathrm{e}^{\mathrm{j}(\omega-\pi)}.$$

Use equation (9.18) at this shifted frequency. Since L is odd,

$$\begin{aligned} P_0(-\mathrm{e}^{\mathrm{j}\omega}) &= \mathrm{e}^{-\mathrm{j}L(\omega-\pi)} P_{\text{ampl}}(\omega-\pi) \\ &= \mathrm{e}^{-\mathrm{j}L\omega} \underbrace{\mathrm{e}^{\mathrm{j}L\pi}}_{-1} P_{\text{ampl}}(\omega-\pi) \\ &= -\mathrm{e}^{-\mathrm{j}L\omega} P_{\text{ampl}}(\omega-\pi). \end{aligned}$$

Consequently the *difference* of the two product responses becomes a *sum* of the two signed amplitudes:

$$P_0(\mathrm{e}^{\mathrm{j}\omega}) - P_0(-\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}L\omega}\big[P_{\text{ampl}}(\omega) + P_{\text{ampl}}(\omega-\pi)\big].$$

Cancel the nonzero phase factor in equation (9.10) to obtain

$$\boxed{P_{\text{ampl}}(\omega) + P_{\text{ampl}}(\omega-\pi) = 2.} \tag{9.20}$$

Because the amplitude is even, $P_{\text{ampl}}(\omega-\pi) = P_{\text{ampl}}(\pi-\omega)$. On $[0,\pi]$, the two graphs are reflections about $\pi/2$.

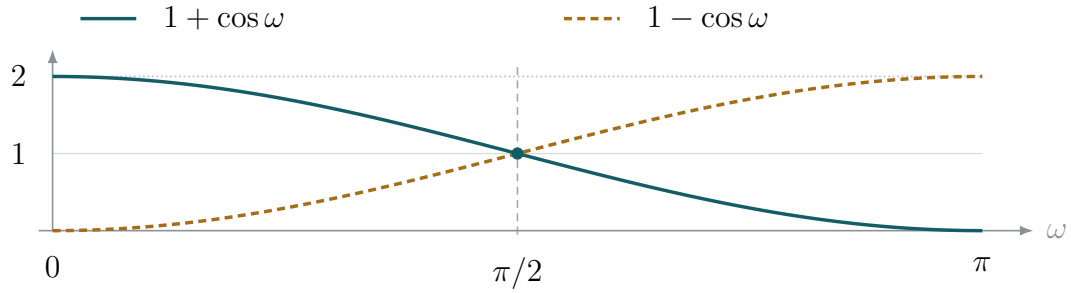


Figure 9.9: The three-tap amplitude and its shifted copy. Their sum is exactly two at every frequency, not just in the passband and stopband.

At the midpoint, the two values agree, so $P_{\text{ampl}}(\pi/2) = 1$. At the endpoints, $P_{\text{ampl}}(0) + P_{\text{ampl}}(\pi) = 2$. For a low-pass product, we want the first endpoint value near two and the second near zero.

This is not an arbitrary functional design problem: an FIR Type-I product has the *finite cosine-polynomial* form in equation (9.19). A general complementary function need not correspond to an FIR filter.

9.12 Passbands, stopbands, and matched ripples

Let $\omega_p < \pi/2$ be a passband edge and $\omega_s > \pi/2$ a stopband edge. The desired signed amplitude is near two on $[0, \omega_p]$ and near zero on $[\omega_s, \pi]$. To pair these bands under the reflection $\omega \mapsto \pi - \omega$, choose

$$\boxed{\omega_p + \omega_s = \pi, \qquad \omega_s - \frac{\pi}{2} = \frac{\pi}{2} - \omega_p.} \tag{9.21}$$

These are symmetric specifications around half the Nyquist frequency. They do not, by themselves, guarantee perfect reconstruction: the full amplitude identity must hold at *every* frequency, including the transition band.

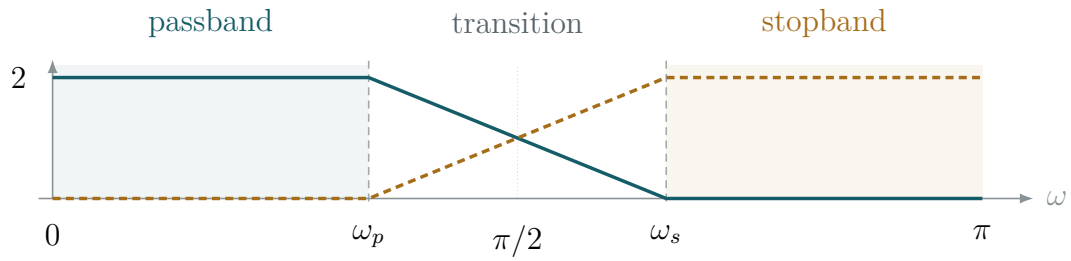


Figure 9.10: Complementary amplitude specifications. Reflection exchanges the passband and stopband; the two band edges are symmetric about $\pi/2$.

Ripples cannot be chosen independently. If a passband value is

$$P_{\text{ampl}}(\omega) = 2 + \varepsilon(\omega),$$

then equation (9.20) forces its paired stopband value to be

$$P_{\text{ampl}}(\pi - \omega) = -\varepsilon(\omega).$$

A positive passband error requires an equally large negative stopband error. The signed errors cancel; their magnitudes do not.

9.13 Build complementarity into the FIR coefficients

Although the amplitude identity must hold at every frequency, it can be enforced directly through a finite set of coefficients. The parity restriction obtained earlier removes exactly the cosine terms that would violate complementarity.

For a symmetric product centered at odd L, we already know $p_0[L] = 1$. If $r > 0$ is even, then $L - r$ is an odd index different from L, so $p_0[L - r] = 0$. The cosine expansion therefore contains only odd harmonics:

$$\boxed{P_{\text{ampl}}(\omega) = 1 + \sum_{\substack{1 \le r \le L \\ r \text{ odd}}} b_r \cos(r\omega), \qquad b_r = 2p_0[L - r].} \tag{9.22}$$

Conversely, choose real coefficients b_r in this finite sum and set $p_0[L \pm r] = b_r/2$. Each odd harmonic changes sign after a π shift:

$$\cos\big(r(\omega - \pi)\big) = (-1)^r \cos(r\omega) = -\cos(r\omega).$$

Thus all the cosine terms cancel in the sum of the shifted amplitudes, leaving two. The product identity is automatic. We can select the free coefficients to improve the low-pass shape without losing that identity.

A concrete ripple illustration. Consider the following finite cosine polynomial. It is useful for illustrating complementarity, although its coefficients have not been optimized for a particular passband or stopband tolerance:

$$R(\omega) = 1 + \frac{4}{\pi} \sum_{r=0}^{4} \frac{(-1)^r}{2r+1} \cos\big((2r+1)\omega\big).$$

It has odd harmonics only, so $R(\omega) + R(\omega - \pi) = 2$ exactly.

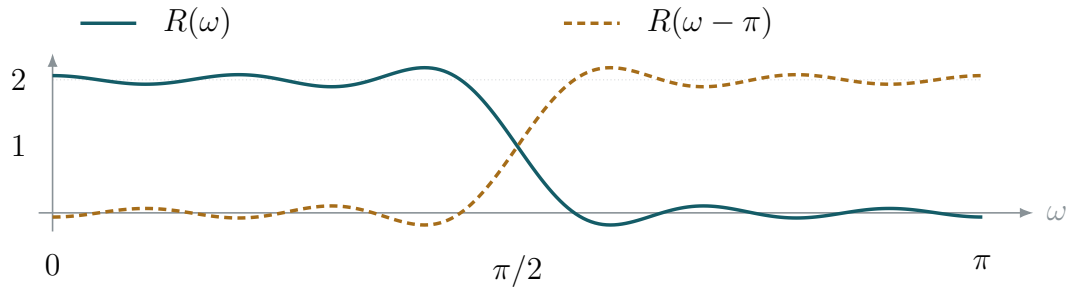


Figure 9.11: An actual finite cosine-polynomial pair. The ripples are signed and cancel exactly, including wherever one amplitude is negative.

The corresponding symmetric FIR product has center $L = 9$, center tap one, and

$$p_0[9 \pm (2r+1)] = \frac{2(-1)^r}{\pi(2r+1)}, \qquad r = 0, \ldots, 4.$$

All other taps are zero. This supplies a direct route from a finite frequency-domain choice back to an implementable product filter.

9.14 Ideal low-pass factors and the gain of two

The final example is the limiting picture in which the transition band disappears. On $[-\pi, \pi]$, set

$$C(\omega) = \mathbf{1}_{(-\pi/2, \pi/2)}(\omega),$$

and extend C periodically. Away from the band edges, $C(\omega) + C(\omega - \pi) = 1$. Values at those isolated edges do not affect the inverse DTFT.

The ideal *signed amplitude* is $P_{\text{ampl}}(\omega) = 2C(\omega)$. To retain our odd-delay convention, the full product response is

$$P_0(\mathrm{e}^{\mathrm{j}\omega}) = 2\mathrm{e}^{-\mathrm{j}L\omega}C(\omega), \qquad L \text{ odd}. \tag{9.23}$$

The rectangle depicts the signed amplitude only. The delay factor remains part of the full frequency response and is needed to satisfy the product identity with the stated indexing. Omitting it would change the reconstruction convention.

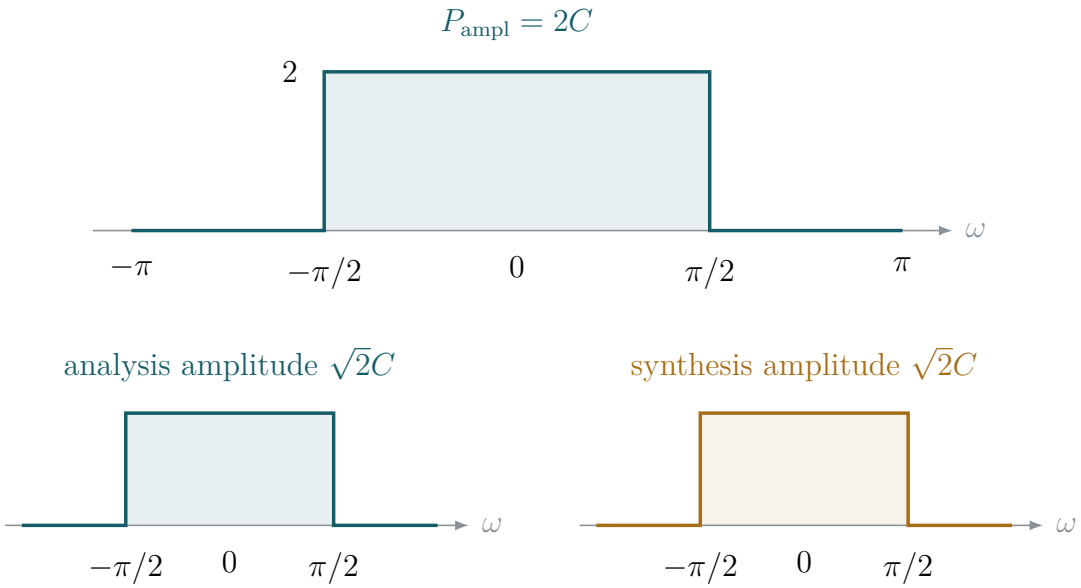


Figure 9.12: The ideal amplitude and an equal-gain factorization. Phase factors are not shown in these amplitude plots; their delays must add to L.

The gain can be split between the factors. Away from the band edges, choose any $\alpha > 0$ and integer shifts L_h, L_g with $L_h + L_g = L$:

$$H_0(\mathrm{e}^{\mathrm{j}\omega}) = \alpha\mathrm{e}^{-\mathrm{j}L_h\omega}C(\omega), \qquad G_0(\mathrm{e}^{\mathrm{j}\omega}) = \frac{2}{\alpha}\mathrm{e}^{-\mathrm{j}L_g\omega}C(\omega).$$

Since $C^2 = C$, their product is equation (9.23). The gain pairs $(2, 1)$, $(\sqrt{2}, \sqrt{2})$, and $(1, 2)$ all have product two. The complementary-band identities hold away from the band edges.

Where does the factor two come from? Recall that downsampling followed by zero insertion retains the even positions:

$$(\uparrow 2 \downarrow 2\, v)[n] = \frac{1 + (-1)^n}{2}\, v[n].$$

Its z-transform is $\frac{1}{2}[V(z) + V(-z)]$, which is the source of the $1/2$ in equation (9.1). The total desired-channel response must therefore be $2z^{-L}$, not z^{-L}, to produce unit reconstruction gain.

This does *not* mean that downsampling divides every retained sample by two: $(\downarrow 2\, v)[m] = v[2m]$. The factor $1/2$ appears in the spectral decomposition of the even-index selector.

The ideal product is not FIR. Inverting equation (9.23) makes this explicit:

$$\begin{aligned} p_0[n] &= \frac{1}{2\pi}\int_{-\pi}^{\pi} P_0(\mathrm{e}^{\mathrm{j}\omega})\mathrm{e}^{\mathrm{j}\omega n}\,\mathrm{d}\omega \\ &= \frac{1}{\pi}\int_{-\pi/2}^{\pi/2} \mathrm{e}^{\mathrm{j}\omega(n-L)}\,\mathrm{d}\omega. \end{aligned}$$

For $n \neq L$, evaluate the exponential integral:

$$p_0[n] = \frac{1}{\pi}\frac{\mathrm{e}^{\mathrm{j}\pi(n-L)/2} - \mathrm{e}^{-\mathrm{j}\pi(n-L)/2}}{\mathrm{j}(n-L)} = \frac{2\sin\big(\pi(n-L)/2\big)}{\pi(n-L)}.$$

At $n = L$, the integral is one. Thus

$$\boxed{p_0[n] = \operatorname{sinc}\left(\frac{n-L}{2}\right), \qquad \operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t}, \quad \operatorname{sinc}(0) = 1.} \tag{9.24}$$

It satisfies $p_0[2m + L] = \operatorname{sinc}(m) = \delta[m]$, but has infinitely many nonzero taps on the other parity. The ideal frequency picture is therefore a design target, not an FIR realization.

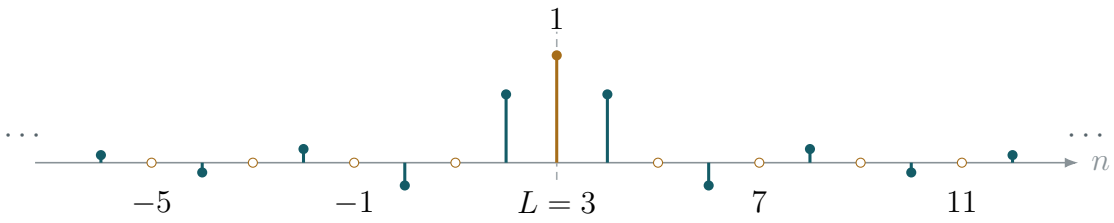


Figure 9.13: The ideal product impulse response for $L = 3$. Odd-indexed taps vanish except at the center, while the even-indexed sinc tail continues indefinitely.

We have reduced perfect reconstruction to a condition on the product filter and identified how the remaining coefficients control its frequency response. The next step is to impose orthogonality on the bank. In Lecture 10, the filtered and downsampled outputs will become coordinates in an orthonormal basis of $\ell^2(\mathbb{Z})$. With appropriate additional conditions, these discrete coordinates will then lead to orthonormal wavelet systems.